

Thisal Mandula Sugathapala

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I am a Ph.D. student in the Division of Fluid Mechanics at Chalmers University of Technology, working on developing multiscale models to track the transport of microplastics in the ocean. My research aim is to develop a framework for environmental assessment that integrates numerical studies of microplastics with broader ecological implications. The goal of my research is to use numerical models to explain observed accumulation patterns of microplastics in the ocean and contribute to the future development of marine policy to mitigate the negative impacts of microplastic pollution.

Education

Ph.D. in Fluid Mechanics (Reading)

Chalmers University of Technology

November 2022 – May 2027

Research Topic: Multi-scale models for the transport of microplastics in a marine environment.



Master's in Mechanical Engineering

Linköping University

August 2020 – September 2022

GPA: 4.8/5

Specialization: Applied Mechanics (Fluid Mechanics Stream)

Scholarships: Awarded a LiU International Scholarship.



Bachelor's in Mechanical Engineering

Sri Lanka Institute of Information Technology

January 2015 – November 2018

Senior Year GPA: 3.51/4 (Rank 1)

Graduated with Second Class Honors (Upper Division)

Graduated 1st in Class: Recipient of the Best Performance Award for the BSc (Hons) in Mechanical Engineering, 2019.



Conferences (Oral Present.)

12th International Conference on Multiphase Flow.

12th to 16th May, 2025, Toulouse, France.

76th Annual Meeting of the Division of Fluid Mechanics.

19th to 21st November, 2023, Washington DC, USA.

1st International Conference on Green Energy and Environmental Technology.

24th to 26th July, 2019, Paris, France.

Publications

Thisal Sugathapala

Multi-scale modeling of microplastic transport in marine environments

Chalmers Publication Library, 2025.

<https://research.chalmers.se/publication/548253>

Thisal Sugathapala et al.,

"Vertical transport of buoyant microplastics in the ocean: The role of turbulence and biofouling. Environmental Pollution, 2025.

<https://doi.org/10.1016/j.envpol.2025.125819>

Thisal Sugathapala et al.,

"Co-Simulation Development for Improved Cavitation Predictions in Oil-Hydraulic Systems", LiU University Press, 2022.

liu.diva-portal.org/cavitation

Thisal Sugathapala et al.,

"Simulation Strategies for improved contamination modeling of liquid dynamics on automotive surfaces", LiU University Press, 2022.

liu.diva-portal.org/contamination

Thisal Sugathapala et al.,

"Aerodynamics modeling of simplified wind turbine rotors targeting small-scale applications in Sri Lanka", Energy for Sustainable Development, Elsevier, 2020.

<https://doi.org/10.1016/j.esd.2020.09.004>

Relevant Work Experience

Master Thesis

Volvo Cars and Volvo Trucks

CFD Student

January 2022– June 2022

Carried out my master thesis in collaboration with Volvo Cars and Volvo Trucks for an experimental and numerical study on “Simulation strategies for improved contamination modeling of liquid dynamics on automotive surfaces”.



Research Project

Epiroc AB

CFD Student

August 2021–January 2022

Conducted research on co-simulation development for enhanced cavitation predictions in oil-hydraulic systems, utilizing a multiphase mixture model for oil and oil vapor in collaboration with Epiroc AB.



Linköping University

Liu Formula Student

CFD Engineer

August 2021–August 2022

Worked on full-car CFD simulations on high-performance computers to design and optimize aerodynamic parts for the ER22 formula car, which competed during the summer of 2022 in the FS UK and FS East competitions held in England and Hungary, respectively.



Computational Mechanics

Linköping University

Teaching Assistant

August 2021– June 2022

Worked as a teaching and lab assistant for numerous courses, including Introduction to Computational Mechanics for the Division of Solid Mechanics and Computational Fluid Dynamics, and Computational Aerodynamics for the Division of Applied Thermodynamics and Fluid Mechanics at Linköping University.



Department of Mechanical Engineering

Sri Lanka Institute of Information Technology

Teaching and Research Assistant

February 2019–August 2020

Worked as a teaching and lab assistant for courses on Fluid Mechanics, Thermodynamics, and Renewable Energy.



Industrial Internship

Sri Lanka Sustainable Energy Authority

Engineering Intern

November 2017–January 2018

Worked under the guidance of the Deputy Director General and was involved in the design of a solar monitoring tower, the creation of mathematical models to predict solar panel performance, and the development of a new wind atlas for Sri Lanka. Additionally, organized the National Energy Symposium in 2017.



Skills

Computer Skills:

- Microsoft Office
- Dassault Systems Solidworks
- Autodesk AutoCAD
- MathWorks Matlab
- Fortran
- Python
- OpenFOAM
- ModeFRONTIER
- ANSYS FLUENT
- ANSYS Design Explorer
- Simcenter Star CCM+
- ANSA – Pre-Processor
- ParaView – Post Processor

Language Skills:

- English (Fluent)
- Sinhala (Native)

PhD Courses

	Grade
Artificial Intelligence for Engineering Applications (<i>Summer school at TU Delft, Netherlands, organized by IDEA League</i>)	G
Multiphase Flows	G
GPU-accelerated computational methods using Python and CUDA	5
Towards sustainable shipping	5
Computational Fluid Dynamics Modeling with Open-Source Software. (<i>Course taken at Politecnico di Milano, Italy, as an exchange PhD student</i>)	A

Master's Transcript

	Grade
FIRST YEAR	
Fluid Power Systems	5
Continuum Mechanics	5
Engineering Materials	5
Introduction to Computational Mechanics	4
Product Development	4
FEM: Advanced Course	5
CFD: Basic Course	5
CFD: Advanced Course	5
Aerodynamics: Continued Course	5
Damage Mechanics and Life Analysis	5
SECOND YEAR	
CFD: Applied Course	5
Gas Turbine Engines	4
Computational Heat Transfer	5
CFD: Project Course	5
Master Thesis	G

Grading Scale: Pass with Distinction (5), Pass with Credit (4), Pass (3), Fail (U), Pass with No Grade (G). Course descriptions can be found at: